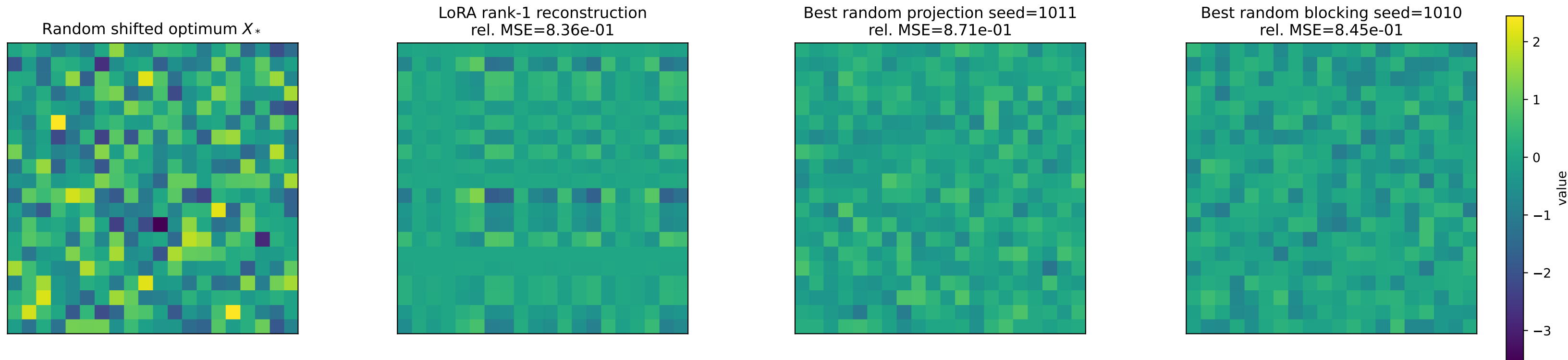


Random full-rank shift exposes LoRA rank limitation ($M = 20$, LoRA rank $r = 1$, matched search dimension $d = 40$)



What this tests

Target is a random Gaussian matrix, which is generally full rank.

Rank- r LoRA no longer contains the true optimum; it can only produce the best rank- r approximation.

The shifted benchmark optimum is still exactly at this random target, so this is a limitation test.

